

LAB REPORT

Title: Functions and Arrays in C++

Objective

- To understand the concept of functions in C++
- To learn how to call functions from main()
- To practice array input, processing, and output
- To calculate sum and average using functions

Apparatus / Software

- Computer System
- C++ Compiler (Code::Blocks / Dev C++ / VS Code)

Theory

Functions in C++ are used to divide a program into smaller blocks to improve readability and reusability. Arrays are used to store multiple values of the same data type. Functions can also be used to perform operations on arrays, such as summing elements or calculating averages.

Program 1: Greeting Function

Code:

```
void greet()
{
    cout << "Welcome to C++ Programming!" << endl;
}

int main()
{
    greet();
    return 0;
}
```

Output:

Welcome to C++ Programming!

Program 2: Hello World Function

Code:

```
void sayHello()
{
```

```
    cout << "Hello, World!" << endl;
}

int main()
{
    sayHello();
    return 0;
}
```

Output:

Hello, World!

Program 3: Sum and Average of Array

Code:

```
int calculateSum(int arr[], int size)
{
    int sum = 0;

    for(int i = 0; i < size; i++)
    {
        sum = sum + arr[i];
    }

    return sum;
}

int main()
{
    int numbers[10];
    int sum;
    double average;

    cout << "Enter 10 integer values:" << endl;

    for(int i = 0; i < 10; i++)
    {
        cin >> numbers[i];
    }

    sum = calculateSum(numbers, 10);
    average = sum / 10.0;

    cout << "Sum = " << sum << endl;
    cout << "Average = " << average << endl;

    return 0;
}
```

Sample Output:

Enter 10 integer values:
1 2 3 4 5 6 7 8 9 10
Sum = 55
Average = 5.5

Conclusion

In this lab, we learned how to use functions in C++ for different tasks such as displaying messages and performing calculations. We also practiced working with arrays to store multiple values and compute their sum and average. This lab helped improve understanding of modular programming and array handling.